



Project Title: CleanEase – Smart Laundry System

Course: Software Design & Architecture (SDA)

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Contents

1. Introduction	3
2. System Features.....	3
3. Design Patterns Used	3
System Architecture	4
Use Case	4
Class Diagram	5
Sequence Diagram (Flow)	6
Package Diagram.....	7
System Sequence Diagrams	8
Advantages	11
Limitations	11
Future Enhancements	11
Conclusion.....	12

1.Introduction

CleanEase is a console-based Laundry Management System that allows users to select laundry services, calculate bills, and receive order status updates. The system is designed using design patterns to demonstrate real-world software design principle

2.System Features

- Customer registration (basic input)
- Laundry service selection
- Price calculation based on service and quantity
- Order processing
- Real-time order status updates
- Notification system

3.Design Patterns Used

i. Factory Pattern (Creational Pattern)

Purpose: Used to create laundry service objects dynamically.

Implementation: The Service-Factory class creates objects like Wash, Iron, and Dry Clean based on user input.

Why Used:

- Avoids direct object creation
- Improves scalability
- Makes system easy to extend

ii. Observer Pattern (Behavioral Pattern)

Purpose: Used to notify customers when order status changes.

Implementation: Customer acts as Observer, and Order acts as Subject.

Why Used:

- Enables automatic notifications
- Reduces dependency between objects
- Improves communication between modules

System Architecture

The system follows a simple object-oriented architecture:

- Factory Pattern → Service creation
- Observer Pattern → Notifications

Use Case



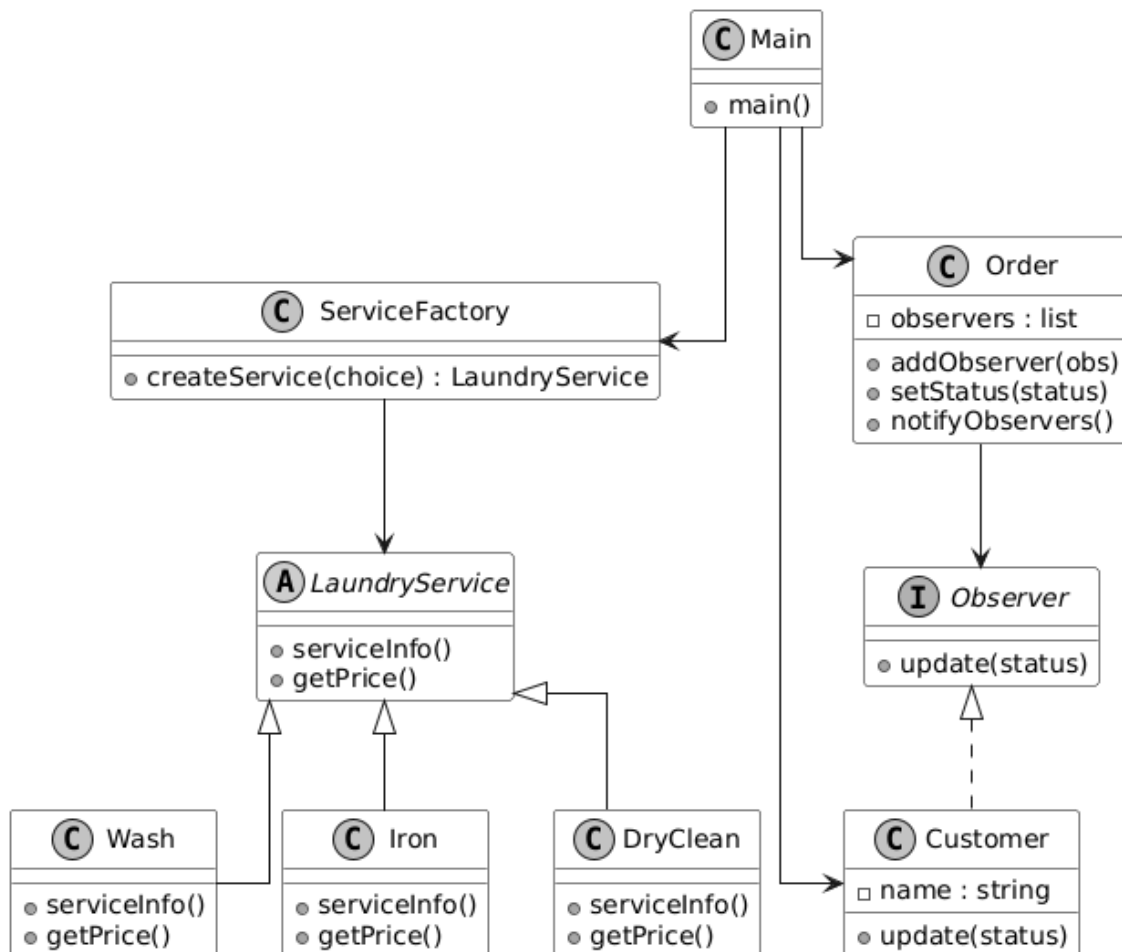
Class Diagram

Classes:

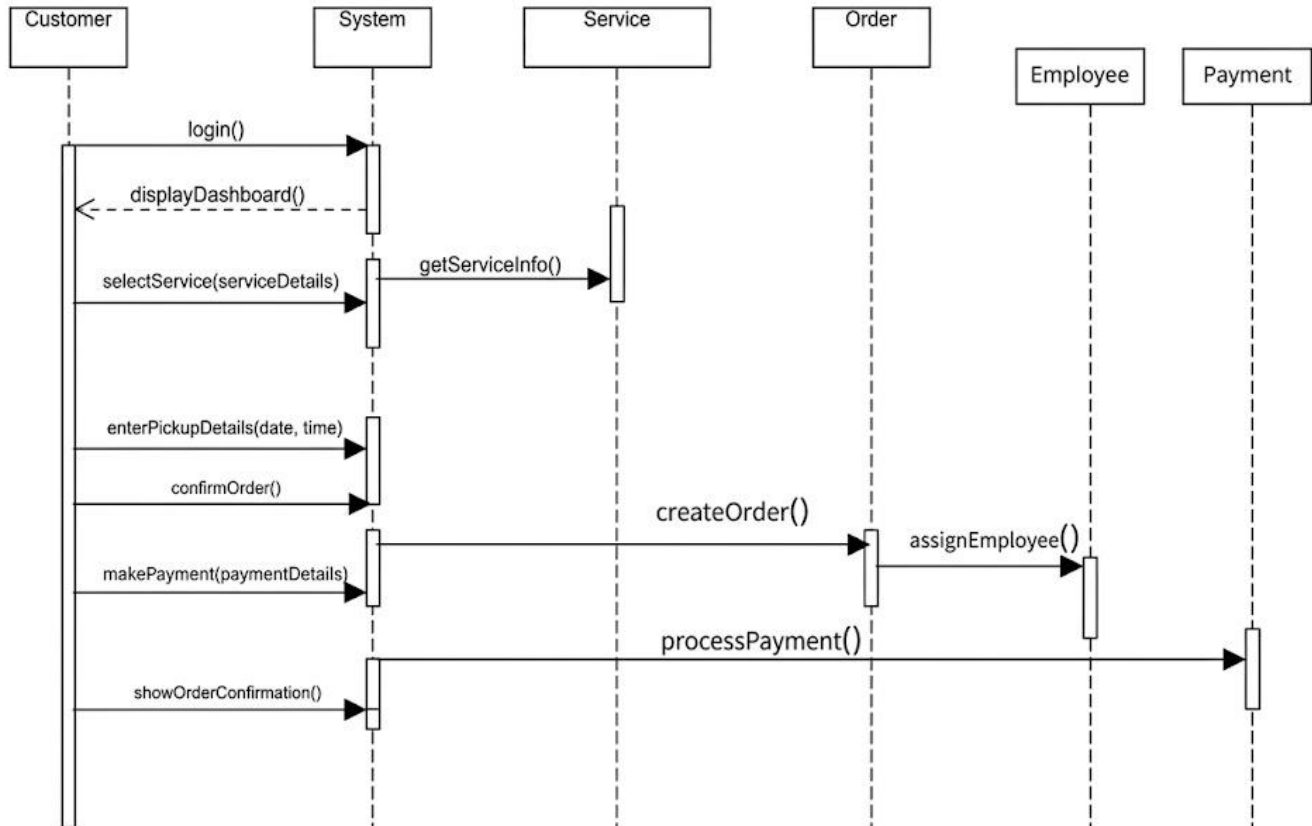
- Laundry-Service (Abstract Class)
- Wash, Iron, Dry-Clean (Concrete Classes)
- Service-Factory
- Observer (Interface)
- Customer
- Order

Relationships:

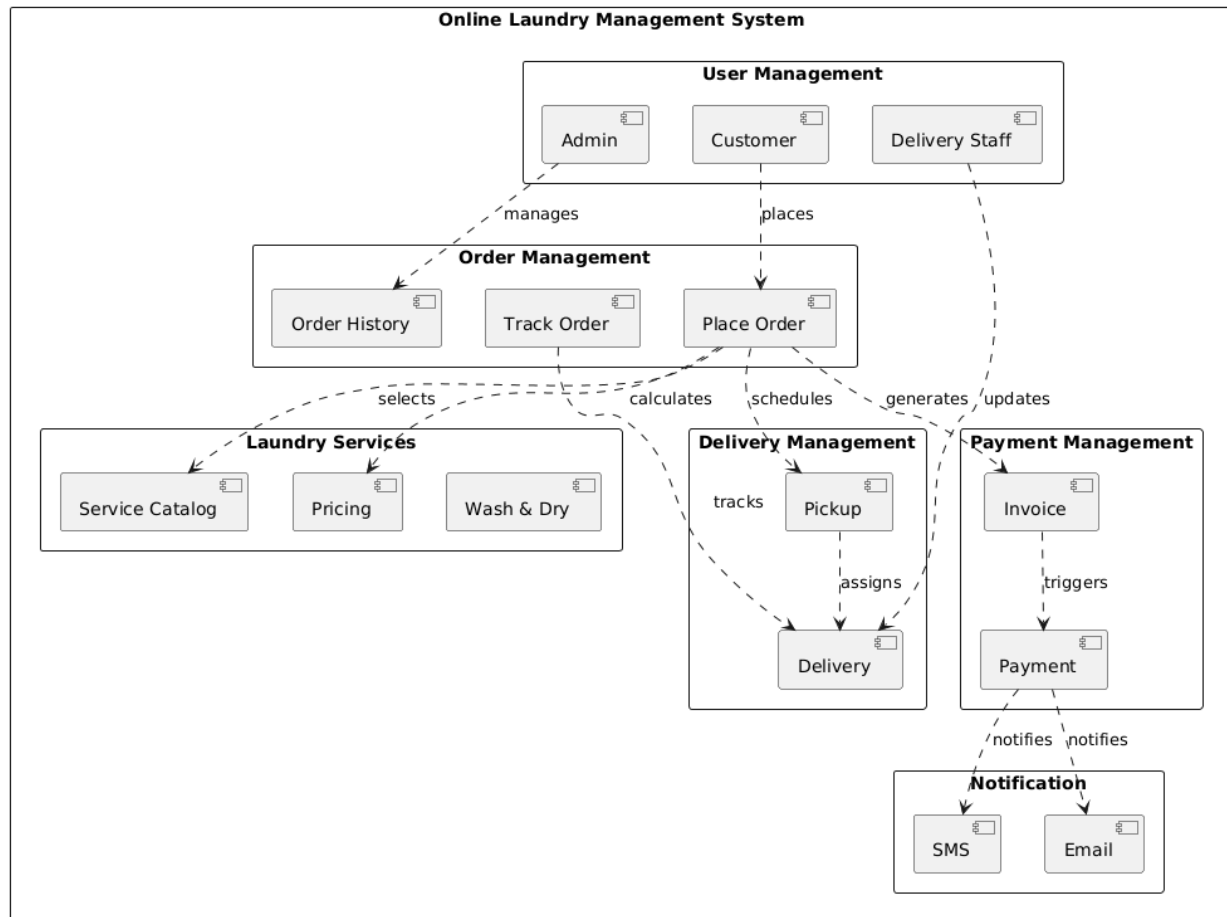
- Inheritance between Laundry-Service and service types
- Observer pattern between Order and Customer



Sequence Diagram (Flow)

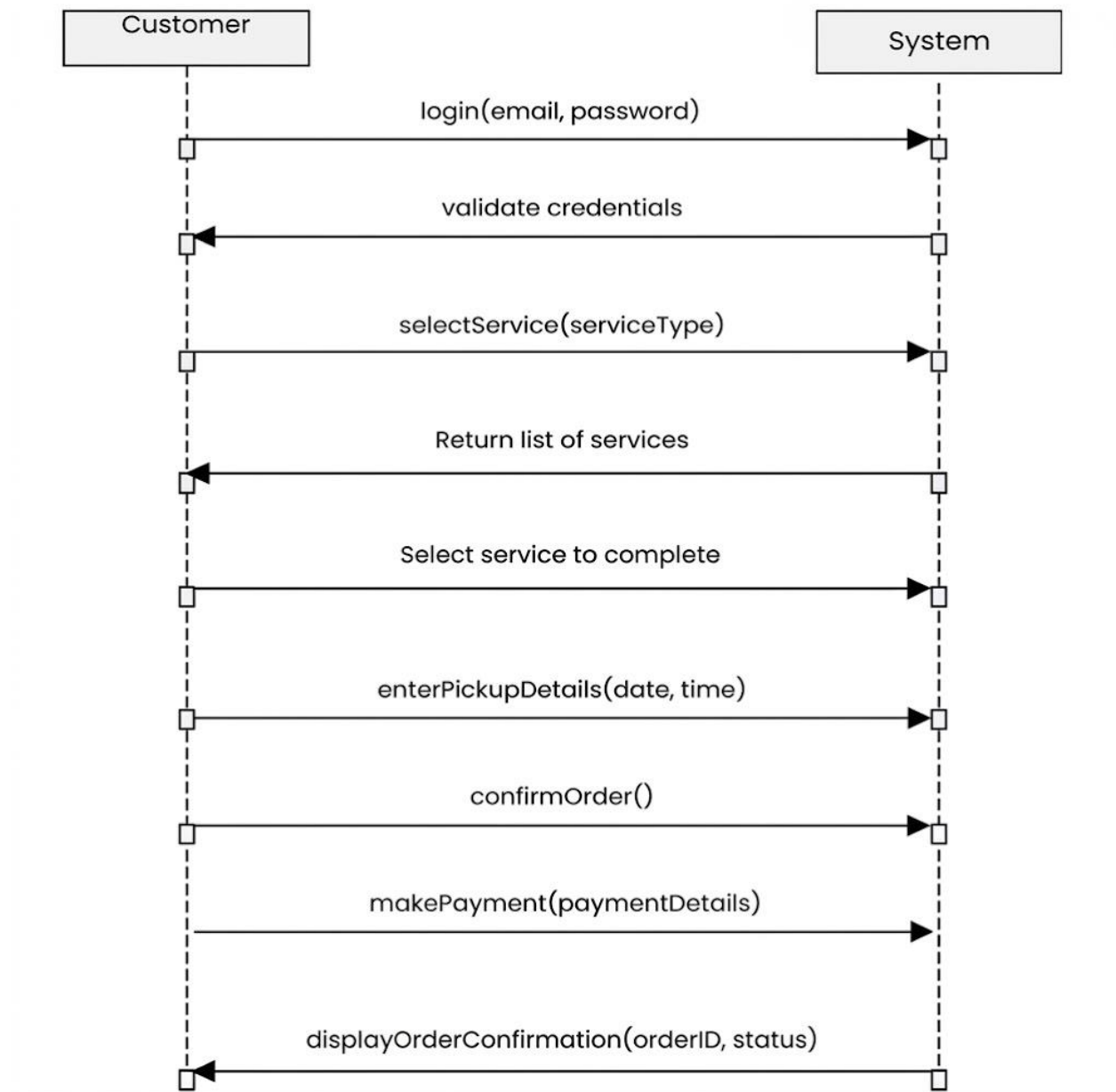


Package Diagram

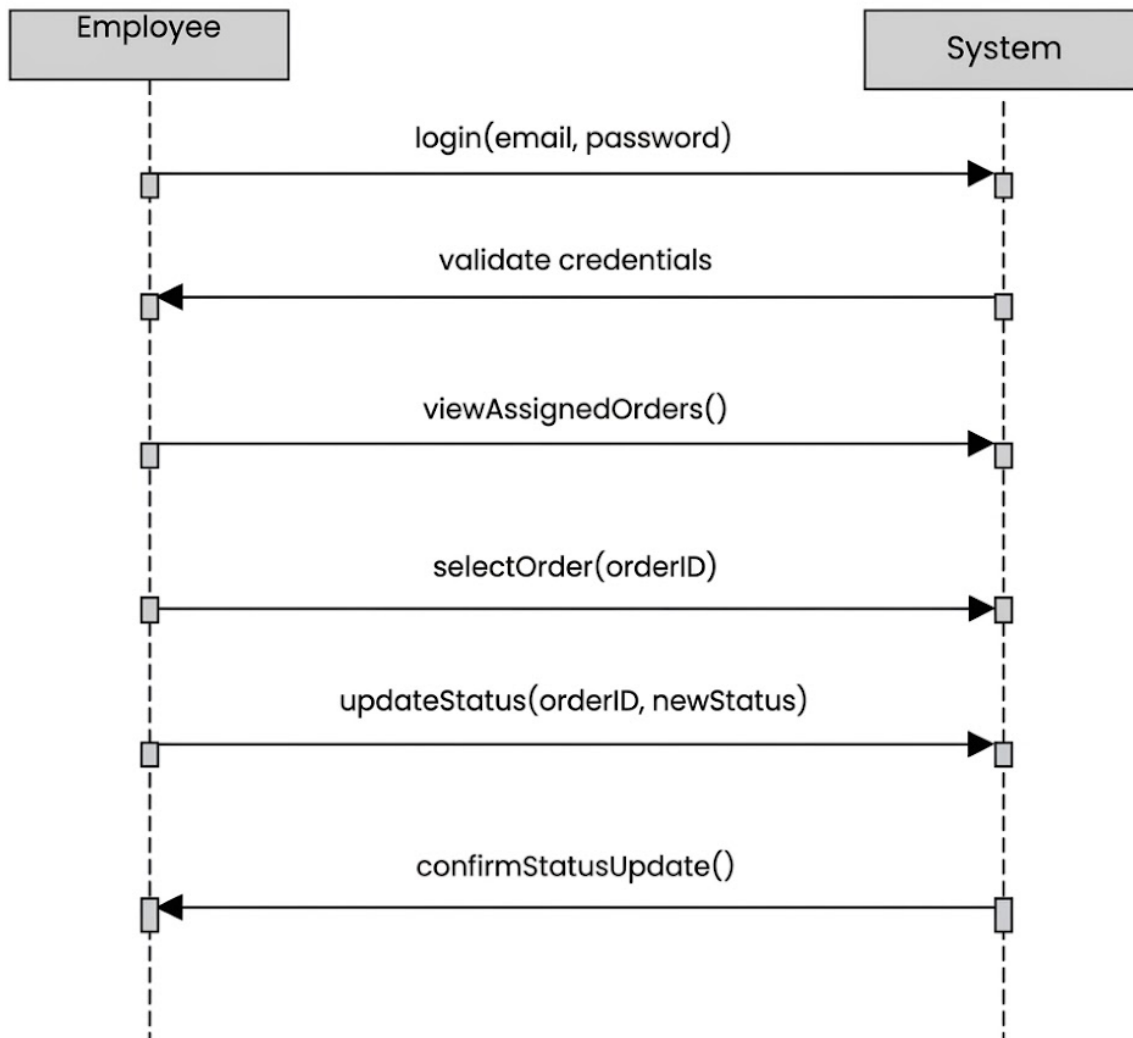


System Sequence Diagrams

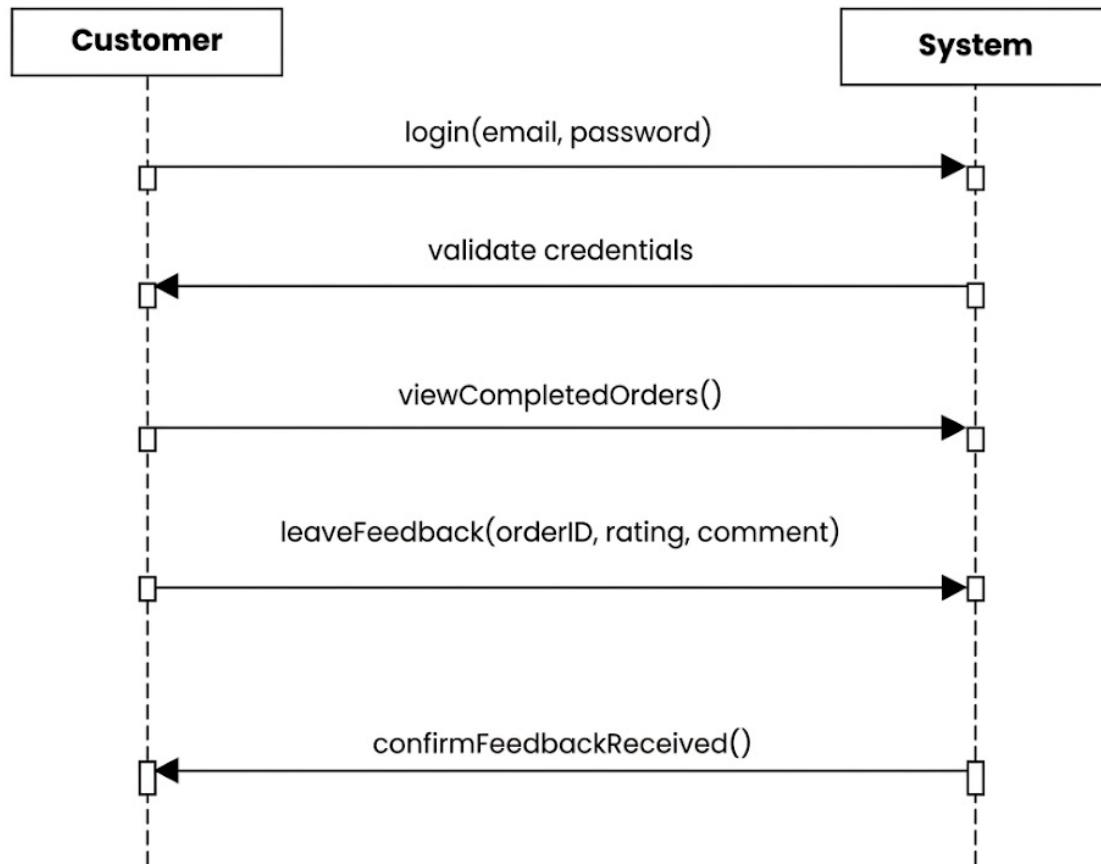
I. Place Laundry Order



II. Track Order



III. Add Feedback



Advantages

- Simple and user-friendly design
- Modular and scalable system
- Easy to add new services
- Real-time notifications
- Demonstrates design patterns effectively

Limitations

- Console-based system (no GUI integration)
- No database storage
- Limited payment simulation

Future Enhancements

- Web-based interface
- Database integration
- Online payment system
- Mobile application
- Delivery tracking system

Outputs

```
Enter Customer Name: Fatima

Select Laundry Service:
1. Wash (Rs.500)
2. Iron (Rs.300)
3. Dry Clean (Rs.800)
Enter Choice: 1
Wash Service Selected

Enter Number of Clothes: 8

Total Bill: Rs.4000

===== Order Updates =====

Notification for Fatima: Order Placed Successfully

Notification for Fatima: Laundry Processing Started

Notification for Fatima: Laundry Completed

Notification for Fatima: Order Ready for Delivery
```

Conclusion

The CleanEase Laundry Management System successfully demonstrates the use of Factory and Observer design patterns in a real-world scenario. It improves code reusability, scalability, and separation of concerns.

